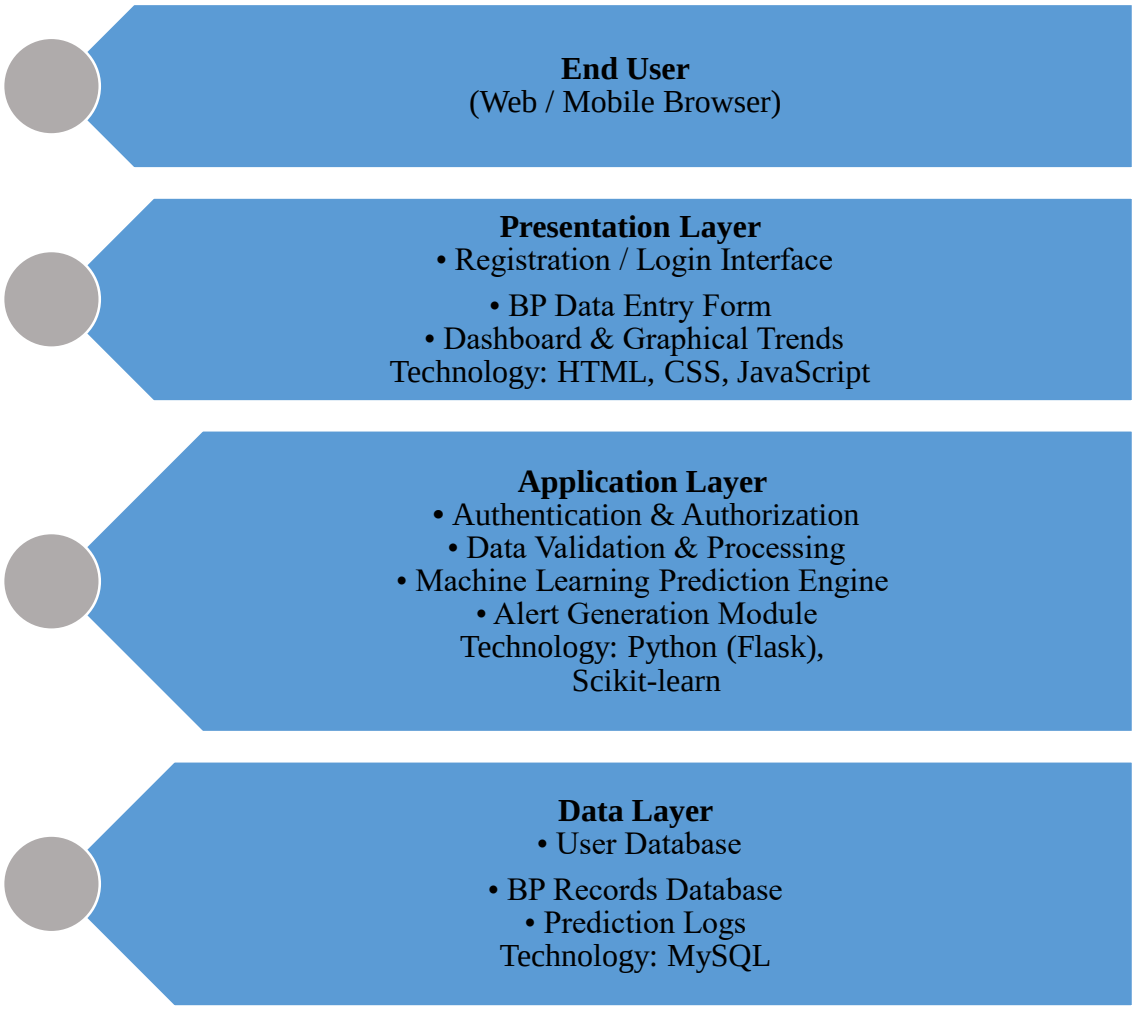
Project Design

Date	19 September 2022
Team ID	
Project Name	Predictive Pulse : Harnessing machine Learning for Blood Pressure
Maximum Marks	2 Marks

Solution Architecture:



- The user accesses the web application through a browser.
- The presentation layer collects user credentials and blood pressure readings.
- The application layer processes input data and sends it to the Machine Learning model for prediction.

- The ML engine analyzes historical and current BP data to determine the risk level.
- The results are stored in the database and displayed on the dashboard with alerts if necessary.

Proposed Solution :

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Many individuals suffer from irregular blood pressure but lack continuous monitoring and early risk prediction systems, leading to delayed diagnosis and health complications.
2.	Idea / Solution description	Develop a Machine Learning-based web application that analyzes blood pressure readings and predicts risk levels (Low/Medium/High), enabling proactive healthcare monitoring.
3.	Novelty / Uniqueness	The system combines real-time BP data tracking with predictive analytics, offering early warning alerts instead of just displaying static readings.
4.	Social Impact / Customer Satisfaction	Helps users monitor cardiovascular health proactively, reduces hospital dependency, and promotes preventive healthcare awareness.
5.	Business Model (Revenue Model)	Freemium model – Basic prediction features free; advanced analytics, detailed reports, and doctor integration available via subscription.
6.	Scalability of the Solution	Designed using 3-tier architecture, allowing cloud deployment and support for increasing users and large datasets without performance degradation.

Problem Solution Fit Analysis

1. CUSTOMER SEGMENTS (CS) CS

- Adults suffering from hypertension
- Individuals at risk of high/irregular plressure
- Health-conscious users seeking proactive health monitoring

2. CUSTOMER CONSTRAINTS (CC) CC

- Lack of time or facilities for regular BP monitoring
- Cost concerns. related to frequent doctor visits
- Fear or anxiety about unexpected high BP readings

4. JOBS-TO-BE-DONE / PROBLEMS (JD) JD

- Need for easy accurate BP tracking
- Timely alerts for health risks
- Lack of early warning systems for critical BP levels

3. AVAILABLE SOLUTIONS (AS) AS

- Manual BP monitors at home for those who own
- Periodic doctor check-ups
- Standard wearable fitness trackers that display current heart rate, but predictive capabilities

5. PROBLEM ROOT CAUSE (RC) RC

- Sporadic BP checking due to busy lifestyle
- Lack of continuous tracking hedl

5. PROBLEM ROOT CAUSE (RC) EB

- Users try to track BP irregularly manual Monitors
- More serious users opt; for periodic doctor visits
- Online/smartphone use for health monitoring

7. TRIGGERS (TR) TR

- Headlines about hypertension risks
- News about lifestyle diseases
- Advice from relatives or friends wI8P issues
- Worsening of existing BP related symptoms

8. YOUR SOLUTION (SL) SL

- Predictive Pulse, a ML-based web app collects BP readings analyizs data, predict risk level (Low/Medium/High) and displays results ontrends with a user-thendly dashooard and analysiaed

9. EMOTIONS: BEFORE / AFTER (EM) EM

- Anxiety & uncertainty. Po, BP risks at warning, (envonsered when uerig sgeen)
- Peace of mind, Pb proactive health insights and early alerts

10. CHANNELS OF BEHAVIOUR (CH) CH

- Predictive Pulse. Web app to input BP rreadings a yeor predictions.
- Online: Doctors for high BP cases based on appn, undergoing regular BP processing